

Civic Lens Implementation Tasks

1. Project Initialization & Foundation

- [x] Await workspace access to C:\Users\Ontop\Desktop\Python_Projects\CivicLens
- [x] Initialize FastAPI backend structure.
- [x] Initialize React (Vite) frontend.

2. Frontend Architecture & PWA Caching

- [x] Install TailwindCSS & Vite PWA plugin.
- [x] Configure Vite Service Workers for deep-offline IndexedDB caching.
- [x] Set up Light/Dark theme provider.
- [] Configure background sync queue (Exponential Backoff).
- [x] Build mobile-first responsive layout shell.

3. Database & Security Foundations

- [x] Integrate PostgreSQL and configure SQLAlchemy ORM.
- [x] Apply TDE (Transparent Data Encryption) configurations.
- [x] Create schema: Locations (State->PU), Users, Results.
- [x] Implement immutable WORM Audit Logs triggers.

4. Authentication & High-Security Access

- [x] Implement JWT Authentication with Argon2 hashing.
- [x] Enforce Multi-Factor Authentication (MFA) for Admins.
- [x] Build Device Fingerprinting logic for field agents.
- [x] Implement **Duress Password (Panic Mode)** distress pipeline.

5. Core API & Data Handling

- [x] Configure strict TLS 1.3 / HSTS network configurations.
- [x] Create cloud image upload pipeline with S3 Server-Side Encryption (SSE-S3).
- [x] Implement SMS Fallback parser & receiver webhook.
- [x] Generate dynamic QR Code for proxy synchronization.

6. Agent Data Capture App (React)

- [x] Build form UI with Soft Geofencing (GPS capture).
- [x] Integrate client-side WASM/JS Image Compression (< 300KB).
- [x] Extract and bundle EXIF Metadata on the client.
- [x] Implement local drafts feature utilizing IndexedDB.

7. Admin Verification & Dashboards

- [x] Build split-screen Verification Dashboard UI.
- [x] Integrate AI / OCR cross-checking logic on the backend.
- [x] Integrate Redis caching for rolling vote aggregations.
- [x] Build interactive choropleth heat-maps & SSE live charts.

8. Agent Provisioning & First-Time Login

- [x] Build Admin endpoint to bulk-create/provision Agent credentials.
- [x] Enforce `must_change_password` flag on new Agent accounts.
- [x] Build Frontend "First-Time Password Reset" screen.

9. User Profile Management

- [x] Design frontend `UserProfileScreen` accessible from main app navigation.
- [x] Display Agent ID, Assigned Polling Unit, and Device Fingerprint status.
- [x] Implement secure endpoint to allow users to update their Main and Duress passwords.

10. Frontend Micro-Architecture Separation

- [x] Scaffold three dedicated Vite environments (`agent-portal`, `admin-portal`, `citizen-portal`).
- [x] Migrate `ResultCaptureForm`, `SyncManager`, and `LoginScreen` to `agent-portal`.
- [x] Migrate `AdminDashboard` and `VerificationScreen` to `admin-portal`.
- [x] Build public `PublicDashboard` in `citizen-portal` stripped of all auth/admin components.
- [x] Cleanup monolithic `frontend` directory.

11. Advanced Security Hardening

- [x] Implement Redis-based **Rate Limiting** on Auth & Verification endpoints.
- [x] Configure **Audit Log Automations** for all sensitive state changes.
- [x] Implement **Admin TOTP (2FA)** for situational room access.
- [x] Add **Client-Side Payload Signing** (HMAC/RSA) to prevent interception tampering.
- [x] Deploy **Security Headers Middleware** (HSTS, CSP, X-Frame-Options).
- [x] Implement **Remote Device Kill-Switch** (token revocation).